Model Verification

Section 4: Risk Assessment and Analyses
The this section will identify the areas of risk in the design, update any modeling or analysis performed to quantify function in the high risk areas, provide adequate justification that the design meets requirements, and serve as an introduction to the prototype test plan

1. Provide a prioritized list of risks derived from a basic Design Failure Modes And Effects Analysis (DFMEA) performed on your design (~ typically 5-10 items in a table – maximum ½ page). A high-level (20 – 30 line, although more are acceptable) overall DFMEA should referenced in this text and documented in an appendix.
    a. ***Note: the following information is not part of the Report 3 template, but is provided for reference on the plan for DFMEA training:***
        - Patrick Gibson will post 2 videos to canvas by April 1, which demonstrate the DFMEA process using a UC Davis ink pen as an example.
        - Patrick Gibson will schedule a 50 minute in-person DFMEA training session the week of April 4th. The date, time an location will be posted to Canvas.
        - Patrick Gibson will also schedule 1 hour follow up Q&A / help sessions on Zoom the week of April 11th. The date, time an location will be posted to Canvas.
        - Patrick Gibson is available for individual team assistance with FMEA by appointment.
2. Document the strategy your team used to manage your highest priority risks. A minimum of 2 risks need to be discussed, but more may be necessary based on the nature of your project and proving that their design meets the major product requirements.
    a. Potential risk management strategies in the design phase are:
        1. Modifying the design to eliminate the failure mode
            - If this is the case, a mini design report detailing the changes and the analysis done to show the new design works as intended is needed.
        2. Performing more detailed analysis using improved research and in-depth well annotated engineering calculations to better quantify the system's performance.
            - Here, results could either confirm proper function of the current design, or could provide evidence that a design change is required and what specific modifications are needed.
        3. Performing more detailed analysis using commercial engineering software
    b. Note: Documentation for each analysis follwing path a2 or a3 above must include the following:
        i. A stated purpose or goal of the analysis
        ii. Assumptions, principles, and methods used (with references where applicable)
        iii. Detailed calculations, simulation settings, and any other information necessary to reproduce the results.
        iv. Results, which should include calculated values, plots, and/or tables.
        v. A discussion detailing how the analysis results may can be interpreted and may impacted the final design, the proposed test plan (next section), or both.

Figure 1.1: *DELETE BEFORE SUBMISSION*

vi. A detailed discussion on the limitations of your selected analyses including: how your assumptions may impact your results, the sensitivity of your analysis to varying inputs, and how you propose to manage these limitations (show the reader that you: "sweated the details", understand the limitations of your analysis, and have considered how this impacts your prototype test plan).

c. Note: the next section (Prototype Test Plan) MUST contain documentation of tests that verify function for the highest priority risk items covered above.

Figure 1.2: *DELETE BEFORE SUBMISSION*

1.0.1 Introduction to Failure Mode Effect Analysis and Methods

The high performance requirements and extreme operating conditions of this device require in-depth risk assessment via Failure Mode Effect Analysis (FMEA). Possible failure modes are identified and ranked by Risk Priority Number (RPN) according to their effect on precision and function of the PTCD as a whole. RPNs are calculated for each failure mode by taking the product of the qualitative ranking of individual factors of failure severity, probability of occurrence, and ease of detection. A critical RPN of **[TODO: critical RPN]** was chosen to be the threshold for in-depth analysis and potential redesign. Note that these failure modes were identified and ranked through preliminary calculations and are verified or challenged by theoretical modeling in Chapter **??**.

[TODO: Insert table of prioritized failure modes (RPN>threshold) from drive. Add full sheet to apendix. It is not finalized yet.]

The threshold RPN was chosen to be **[TODO: critical RPN]**. Since human operators do not come into direct contact or operate the PTCD directly, the main failure risks do not include human injury, but rather device inaccuracy, loss of function, and damage to the PTCD and nearby equipment at the worst case. This RPN value was deemed the critical point at which inaccuracies, loss of function, or damage would prove the device incapable of performing within an acceptable tolerance range **[TODO: if we have time/if possible, see what worst case deformation in critical components can be via transfer function]** Each of the critical failure modes identified must undergo analytical and computational testing to confirm reliability and performance of the current PTCD design, or to inform suggestions for redesign and modification. The results of computational modeling are discussed in **??**

1.0.2 Weld Failure Analysis

Of the critical failure modes **[TODO: number]**? of them are on welds. **[TODO: finish weld calculations, insert calcs in Chapter ??]** The electron beam welding serves to seal the bellows to the housing components, join the housing assembly, and fixture the actuator assembly to the housing. Weld failure at the actuator could cause stroke error, and around the bellows or housing, leakage or critical structural failure could occur. The severity of these methods is reflected in the RPN sheet. To ensure the welds have a sufficient FOS, calculations fed by modeling results were utilized. This process is detailed in Chapter **??**. The housing consists of the inlet and outlet components welded with two central pieces. Together they hold and seal the bellows and orifice assembly in a rectangular space that allows for the actuation movement. The change in geometry and sharp corners flagged this space for potential stress concentrations and region for failure analysis.

The analytical process and results for the above mentioned weld failure methods are documented below:

Housing Center Weld

i. **Purpose.** The purpose of this analysis is to verify that the critical housing weld joints maintain structural integrity under the maximum internal operating pressure of $P = 68.95\,\mathrm{MPa}$, with adequate margin against yielding in both axial tension and bending.

ii. **Assumptions, Principles, and Methods.** The weld joints are treated as square butt joints with full penetration, such that the weld cross-section has the same material properties as the base metal (Invar 36, $S_y = 276\,\mathrm{MPa}$). Because all welds pass 100% radiographic inspection ($E = 1.00$), no weld efficiency penalty is applied [**?** , Table 302.3.4].

   Two load cases are evaluated:

   a. **Axial stress**: The net axial force from internal pressure acting on the wetted area is distributed uniformly across the weld cross-sectional area [**?** , eq. 1-6].

   b. **Bending stress**: The weld is modeled as a beam fixed at both ends, loaded by a uniformly distributed pressure load. While the true geometry is more complex, this conservative beam approximation bounds the bending stress and identifies the key geometric drivers [**?**]. Additional validation using COMSOL Multiphysics is ongoing to account for the complex three-dimensional weld geometry.

   The parameters used in this analysis are summarized in Table **??**.

Table 1.1: *Weld Analysis Parameter Values*

Parameter	Description	Value
P	Internal pressure	68.95 MPa
A_{wetted}	Wetted area (from gas flow analysis)	$3.5504 \cdot 10^{-4}\,\text{m}^2$
A_{weld}	Weld cross-sectional area	$1.3508 \cdot 10^{-3}\,\text{m}^2$
S_y	Yield strength (Invar 36)	276 MPa
t	Weld thickness (full penetration)	7.0 mm

iii. **Calculations.**

Axial Stress. The axial stress is the internal pressure force distributed over the weld area [**?** , eq. 1-6]:

$$\sigma_{\text{axial}} = \frac{P\,A_{\text{wetted}}}{A_{\text{weld}}} = \frac{(68.95 \times 10^6)(3.5504 \times 10^{-4})}{1.3508 \times 10^{-3}} \approx 18.08\,\text{MPa} \tag{1.1}$$

Bending Stress. For a beam of width b and thickness t fixed at both ends under a uniformly distributed pressure load, the maximum bending moment occurs at the fixed ends and at midspan:

$$M_{\text{max}} = \frac{WL^2}{24} = \frac{(Pb)\,L^2}{24} \tag{1.2}$$

Applying the flexure formula with $c = t/2$ and $I = \frac{1}{12}bt^3$:

$$\sigma_{\text{bending}} = \frac{M_{\text{max}}\,c}{I} = \frac{M_{\text{max}}\,(t/2)}{\frac{1}{12}bt^3} = \frac{6\,M_{\text{max}}}{bt^2} = \frac{PL^2}{4t^2} \tag{1.3}$$

Equation **??** shows that bending stress is minimized by reducing the unsupported span L and/or increasing the weld thickness t, providing clear geometric design guidance independent of weld width b.

iv. **Results.**

The axial factor of safety is:

$$FS_{\text{axial}} = \frac{S_y}{\sigma_{\text{axial}}} = \frac{276}{18.08} \approx 15.27 \tag{1.4}$$

The bending factor of safety is:

[TODO: Add data from COMSOL to support or cite where its located in the report]

v. **Discussion.** The axial factor of safety of 15.27 confirms that the weld cross-section carries the pressure-induced axial load with substantial margin, and that axial tension is not a limiting failure mode for this joint. The large safety factor is consistent with the design intent: the weld thickness $t = 7.0\,\mathrm{mm}$ was sized to match the housing wall rather than to minimize material, so the full weld area is available to resist the relatively modest net axial force.

   Bending stress is the more complex and potentially more critical load case. Equation **??** confirms that bending is controlled by the geometric ratio L^2/t^2; keeping spans short and walls thick is therefore the primary mitigation strategy already embedded in the design. However, the beam model assumes a simplified planar geometry that cannot capture the three-dimensional stress distribution at the curved housing interfaces. COMSOL Multiphysics modeling of the full weld joint geometry verifies that the design maintains an adequate factor of safety against bending failure.

   [TODO: Replace bending stress placeholder with COMSOL results and updated safety factor once simulation is complete.]

Actuator-Housing Weldment

i. In addition to stresses applied by the internal pressure, there is also stress on the actuator mount and its' radial weld due to thermal expansion and contraction. This was identified as a potential failure mode for this weld specifically as the other welds are invar-invar and have predictable mechanical properties and a relatively low expansion rate constant across all members in the weldment.

ii. **Purpose.** The purpose of this analysis is to verify that welding is a viable method for joining the actuator subsystem to the housing. Thermal contraction of the actuator mount combined with potential weld embrittlement could cause failure. Further analysis is needed to inform weld reliability or alternate suggestions

iii. **Assumptions, Principles, and Methods.** The weld joints are treated as full penetration fillet welds, and the same assumptions made for the housing weld analysis hold for the material properties and radiographic inspection of the actuator mount to housing weld.

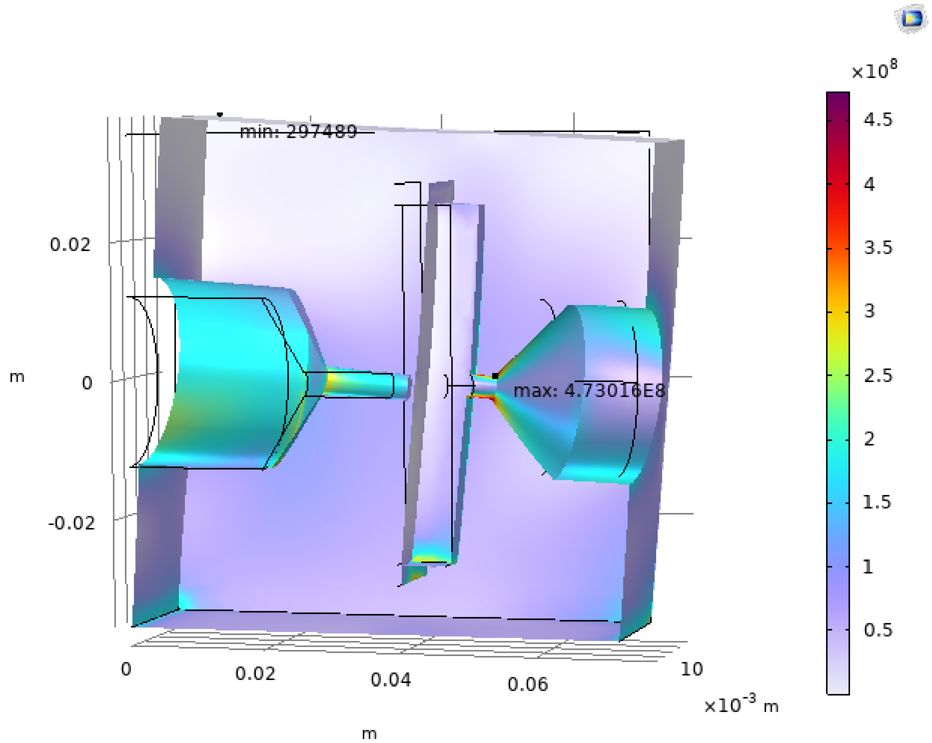

Figure 1.3: *Static FEA Results on simplified Single Body Model*

iv. **Calculations.** The mount being made of Allvar has an anisotropic quality. It has a positive axial coefficient of thermal expansion that is 2 μm/m greater than the magnitude of the negative axial expansion amounting to 32 μm/m. The greatest change in radius ΔR occurs across the cold side of the temperature gradient with $\Delta R = D_0(1 - a_(alv) * dT_cold/2)$ $\Delta R = R_0\,(1 - (-\alpha_{\text{alv}})\,\Delta T_{\text{cold}}$ This is a basic confined thermal expansion problem and the maximum stress on the Allvar and the weld due to the thermal expansion can be calculated by finding the stress necessary to deform the Allvar mount by that amount: $\sigma_{\text{weld}} = \frac{E_{\text{alvar}} \cdot \Delta R}{R_0} = \frac{55 \times 10^9 \cdot 0.0254 \text{ mm}}{18 \text{ mm}} = 74.86$ MPa

v. **Results.** The area of the weld and the area of the mount in contact with it are the same, and thus the stresses are the same. Given that this stress is much smaller than the yield stress of Alvar and of the weld metal we find a sufficient safety factor: $\text{FOS} = \frac{S_{y,\text{weld}}}{\sigma_{\text{weld}}}$

vi. **Discussion.** While this FOS is acceptable, this is assuming the weld has the same material properties as the invar housing. In reality, the material mismatch in the mount-housing weld may lead to detrimental changes in material properties around the weld area which could cause a variety of failures. Because Allvar alloy 30 is a experimental alloy with a proprietary composition we cannot verify the feasibility of this weld. Representatives from the Allvar team advised against welding Allvar to invar or other iron alloys as the titanium from the Allvar will combine with the iron to create brittle intermetallic compounds making the weld brittle. Methods to counter this include using a ductile interface layer that can aid in mitigating creation of brittle compounds, but this could be highly involved and experimental. A recommended path would be to use fasteners with a slight mount redesign that allows for bolting into the housing. This would require some additional research into sealing but is the simplest and most reliable solution.

[TODO: Add in following (purpose, assumpts/principles, results, discussion for all DFMEA topics: housing thin wall, orifice, outlet channel)]

Actuator-Housing Weldament

i. **Purpose.**

ii. **Assumptions, Principles, and Methods.**

iii. **Calculations.**

iv. **Results.** FOS $= \frac{S_{y,\text{weld}}}{\sigma_{\text{weld}}}$

v. **Discussion.**

Testing

[TODO: Outline for Testing chapter]

2.1 Objective

Validate and expand/iterate on analytical methods by simulating the entire PTCD. Testing includes conducting Heat Transfer, Computational Fluid Dynamics (CFD), and Multibody Dynamics (MBD) in isolation, as well as combined.

2.2 Test Design

[TODO: Add corresponding UI screenshots of COMSOL that reflect the boundary condition decisions. Include the COMSOL tree and the actual corresponding part of the model.]

2.2.1 Set-up: Boundary Condition Decisions

- Surface pairs and contact constraints.
- Fluid-solid boundaries (incorporating union nodes where a single continuous boundary is required for FSI).
- Accurate boundary representations for complex orifice shapes, ensuring internal flow geometries are not oversimplified as standard cylinders.

2.2.2 Deformation and Dimensional Tolerance Flow Chart

Mapping the transfer function dependencies: CFD $\rightarrow$ MBD $\rightarrow$ Deformation $\leftarrow$ Heat Transfer.

2.2.3 Mesh Decisions

Strategies and refinements to improve computational accuracy and validity across domains.

2.3 Data Collection

2.3.1 Design Space Matrix

- Turbulent flow ΔP range.
- Heat transfer: T_{amb} range and T_{gas} range.

2.4 Data Analysis

2.4.1 Dimensional Failure

Focusing on heat transfer impacts and thermal expansion.

[TODO: Show images of the failure modes reflected on the model]

2.4.2 Stress and Material Failure

[TODO: Show images of the failure modes reflected on the model. An optional suggestion to have some sort of validation reference via a figure from a study, etc.]

Evaluating structural integrity, yielding, and mechanical limits.